

Shahad Abir

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Professional Summary

Motivated Computer Science undergraduate at CUET with strong problem-solving skills and a proven ability to learn new technologies quickly. Passionate about Computer Vision, Machine Learning, and Software Engineering. Experienced in developing AI-powered applications, research-oriented projects, and scalable full-stack systems. Proficient in building modern web applications and real-time systems using React, Node.js, and Socket.io, with a strong foundation in designing efficient, user-centric software solutions.

Education

B.Sc. in Computer Science & Engineering | Chittagong University of Engineering & Technology (CUET), Chattogram | Jan 2023 – Present

Higher Secondary Certificate (HSC) | Gov. Hazi Muhammad Mohsin College | 2021 | GPA: 5.00/5.00

Secondary School Certificate (SSC) | Khaja Ajmery High School | 2019 | GPA: 5.00/5.00

Technical Skills

Languages: C, C++, JavaScript, Python, Shell scripting (bash)

Frontend: React, React Native, HTML, CSS

Backend & Real-time: Node.js, Express.js, Socket.io, REST APIs

Databases: MySQL, MongoDB (NoSQL), Appwrite Database

DevOps & Microservices: Docker, Nginx, CI/CD Pipelines, Cloud Deployment, Microservices Architecture

Monitoring & Observability: Prometheus, Grafana, Elastic Stack (ELK), Kibana, Jaeger

ML / Data: NumPy, pandas, scikit-learn (classification & regression), PyTorch

Tools: Git, GitHub, Linux

Other: Problem solving, debugging, agile development

Relevant Coursework

- Data Structures & Algorithms.
- Object-Oriented Programming.
- Database Management Systems.
- Operating Systems.
- Computer Networks.
- Software Engineering.
- Machine Learning.

Projects

Diffusion Model Fine-Tuning for Image-to-Image Artwork Generation PyTorch, Diffusers
Fine-tuned a pretrained DDPM UNet on a custom butterfly dataset with DDIM sampling, gradient accumulation, and torchvision preprocessing.

DCGAN Face Generation (CelebA) PyTorch, GANs
Built a DCGAN trained on 200k+ CelebA images and improved training stability to reduce mode collapse.

Video Action Recognition (UCF101) PyTorch, R(2+1)D, OpenCV
Implemented transfer learning from Kinetics-400 using an R(2+1)D CNN for action recognition.

Heart Disease Prediction System Python, scikit-learn
Compared multiple ML classifiers on clinical data, achieving 85–90% prediction accuracy.

Full-Stack E-Commerce Website React, Node.js, MySQL
Developed REST APIs, shopping cart, checkout, and SSLCommerz payment integration using Express and MySQL.

Social Media Web App React, Appwrite, TanStack Query
Built authentication, post creation, infinite scrolling feeds, and efficient frontend caching with Appwrite.

Real-Time Messaging App React, Node.js, Socket.io, MongoDB
Developed a robust real-time 1:1 and group chat with real world messaging features using Socket.io with MongoDB-backed messaging.

Achievements

- Solved 350+ algorithmic problems.
- Hackathon finalist; participated in a Microservices Architecture hackathon.
- Secured 1st place in the Multi-Level Political Meme Classification shared task (DravidianLangTech @ ACL 2026).

Extracurricular Activities

- Member, CUET Computer Club.
- Member, IEEE Computer Society CUET Student Branch Chapter.
- Member, Lions Club Chittagong